



In-situ constructing robust cellulose nanocomposite hydrogel network with well-dispersed dual catalysts for the efficient, stable and recyclable photo-Fenton degradation

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Abstract The hydrogel supported catalyst is explored as an efficient strategy for water remediation due to its combination of absorption and degradation, while it remains great challenge to prepare highly dispersed multi-component catalyst. Herein, cellulose-based hydrogel containing dual-component catalysts was fabricated via in-situ synthesis and mineralization, where the cellulose network containing –OH and –COOH provided a good platform for achieving highly dispersed TiO_2 and Fe_3O_4 . To the methylene blue (MB) solution ($c_0 = 20 \text{ mg/L}$), the $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel displayed a high removal ratio of 97.5% in 60 min by photo-Fenton reaction. Interestingly, owing to the ionic hydrogel feature, the degradation efficiency maintained over

97.2% in a broad initial pH range from 4 to 7, And the hydrogel could effectively hold the contaminants with nearly no leakage and retained the removal ratio of 98.0% after 5 degradation cycles. This study provides new insights into the in-situ synthesis of multi-component catalyst in hydrogels for efficient, stable, and safe water treatment.

Keywords In-situ synthesis · Regenerated cellulose · Nanocomposite hydrogel · Photo-Fenton · Water remediation

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Introduction

The past decades have witnessed the rapidly growing concerns to water pollutant coming from electrolysis, electroplating, pesticide, medicine, paint, paper making, printing and dyeing, textiles and other chemical industries, which is a worldwide threat to human health and sustainable development (Camacho et al. 2020; Yuan et al. 2020; Mudhoo et al. 2020). Several approaches including biodegradation, separation, absorption, sedimentation, as well as chemical degradation with the assistance of O_3 , Fenton catalyst or photocatalyst, have been established for contamination removal from industrial wastewater (Yoo et al. 2017; Ge et al. 2018; Pinto et al. 2018; Xiong et al. 2018; Bandara et al. 2020; Chen et al. 2020). Among

them, physical or chemical process is frequently used as primary treatment to quickly remove contaminants with antimicrobial activity, providing favorable conditions for the subsequent biodegradation. However, physical process only transfers the contaminants from wastewater without degradation. The usage of nanoparticles as catalyst for chemical degradation generally leads to secondary pollution and high cost for the removal of catalyst from the treated water (Reijnders 2008; Rajput et al. 2018). To avoid these limitations, an idea combining absorption and chemical degradation via the immobilization of catalyst in a supporting material, has been proposed to effectively restore water resources, where the supporting materials are employed to absorb the targeted pollutants and the supported catalysts degrade the adsorbed pollutants into harmless or easily treatable components (Lu et al. 2018; Dong et al. 2019; Zhuang et al. 2019; Liu et al. 2020). The strategy not only enhances the stability and reusable performance of the supported catalyst, but also solve the recycling problem of nano-scaled catalysts.

Hydrogel, a typical fascinating soft material, has been well demonstrated to be a promising candidate to support catalysts for water remediation due to its unique three-dimensional network with large specific area, superior adsorption capacity, high water retention, ease of reusability, as well as outstanding environmentally benign features (Zhuang et al. 2019; Kong et al. 2020). But, the degradation activity of supported catalysts will compromise to some extent due to the covering effect of hydrogels, which reduces the effective contacting area at the molecular level between catalysts and contaminants. To address this issue, multi-component composite catalyst system is a commonly used pathway to improve the degradation efficiency of supported catalysts. For instance, Zhang et al. reported a multi-component polyacrylamide composite hydrogel for photo-Fenton reaction via loading ferrous oxide (Fe_3O_4) nanoparticles on the reduced graphene oxide sheets, followed by the combination with the polyacrylamide. By introducing electron-rich materials and physical field, the degradation efficiency was improved, exhibiting a 90% removal ratio with 20 min under visible light irradiation for 20 mg/L Rhodamine B (Dong et al. 2018). A nanocomposite hydrogel was in-situ fabricated by physically blending *N,N*-dimethylacrylamide monomer, titanium dioxide (TiO_2) and graphene oxide, in

which TiO_2 nanoparticles were anchored onto the surface of graphene oxide sheets due to the strong electrostatic interaction. As the electron-rich materials, the reduced graphene oxide was beneficial for accelerating the photocatalysis rate (Wu et al. 2016). On the basis of these studies, it should be noted that the dispersibility of catalysts in the hydrogel matrix is an important parameter to determine the degradation efficiency because the aggregation of supported catalysts will significantly decrease specific surface area and the corresponding available catalytic sites. To the best of our knowledge, physically blending and in-situ mineralization by absorbing precursor have been widely reported for the fabrication of single-component catalyst in the supporting materials (Gao et al. 2014; Zhang et al. 2016; Wang et al. 2017, 2020). However, it is still be a great challenge to develop high degradability activity of nanocomposite hydrogels containing highly dispersed multi-component catalysts, originating from their different formation condition, sizes, charge, surface energy as well as complicated synergistic effect for the removal of contaminants.

Cellulose, an almost inexhaustible biomass materials with abundant polar hydroxyl groups, is considered as an alternative candidate for the fabrication of eco-friendly hydrogels, arising from its intriguing performance such as biodegradability, biocompatibility, thermal and chemical stability (Huang et al. 2014, 2015; Yang et al. 2019). The green alkaline/urea solution developed by Zhang et. al. is a new-generation highly efficient solvent to dissolve cellulose for constructing cellulose hydrogels with eminent structure designability by controlling the strong intra- and inter-molecular hydrogen bonding network (Cai and Zhang 2005; Cai et al. 2008). More interestingly, the abundant active polar groups in the alkaline condition provide a good platform for in-situ formation of metal oxide catalysts via sol-gel method (Das et al. 2017; Wang et al. 2018; Li et al. 2020). Inspired by the understanding of the merits of cellulose hydrogels, we proposed a novel strategy in the current study to fabricate cellulose hydrogels containing polar hydroxyl and carboxyl active sites for accommodating titanium dioxide (TiO_2) and ferrous oxide (Fe_3O_4), respectively, where TiO_2 as photocatalyst was in-situ generated from its precursor with the aid of alkali and Fe_3O_4 as Fenton catalyst was obtained via in-situ mineralization (Gao et al. 2014). Both TiO_2 and Fe_3O_4

were highly dispersed and fully contacted each other at the nano-scale level in the cellulose-based nanocomposite hydrogel for synergistic degradation of organic dyes. Compared with single degradation mechanism of photocatalysis or Fenton reaction, the ingenious structural design of dual-component catalysts endowed the as-prepared cellulose-based nanocomposite hydrogels with an impressive photo-Fenton degradation performance; that is, the removal ratio of organic dyes after photo-Fenton degradation for 60 min was as high as 97.5%. This value is highly attractive in the field of catalytic degradation. Moreover, the high photo-Fenton degradation activity was insensitive to pH ranging from 4 to 7, avoiding the heavy use of acid in practice. The as-prepared $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel was stability and reversibility in the photo-Fenton reaction, exhibiting a high removal efficiency of 97.5% for organic dyes after 5 cycles. The dense and chemically stable hydrogen bonding network in cellulose-based nanocomposite hydrogels was conducive to pinning the organic dye contaminants and reducing their leakage during photo-Fenton catalytic degradation. The methodology proposed provides a promising pathway to address the dispersibility of multi-component catalysts in the hydrogel matrix. This is the first report of in-situ synthesis of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite cellulose hydrogels for water remediation with highly efficient, stable and safe features in a broad pH range.

Experimental section

Materials

The cotton linters with a degree of polymerization of 500 ± 50 were purchased from Hubei Jinhuan Co. Ltd. (Xiangfan, China). Sodium carboxymethyl cellulose (CMC), sodium hydroxide (NaOH), urea, tetrabutyl titanate (TBT), triethanolamine (TEA), ferric chloride hexahydrate ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$), ferrous chloride hexahydrate ($\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$) were provided from Kelong Chemical Reagent Factory (Chengdu, China). All chemicals were used as received without further purification.

In-situ synthesis of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel

Sol-gel method was proposed to in-situ synthesize TiO_2 in the cellulose-based nanocomposite hydrogel. First, the pre-hydrolyzed TBT solution was prepared by mixing TBT, TEA and deionized water under stirring for 2 h. The molar ratio of TEA and TBT was set to be 2.0 and the concentration of TBT in aqueous solution was 1.25 mol/L. 200 g of pre-cooled solution of 7 wt% NaOH, 12 wt% urea and 81 wt% water was used to dissolve 5.5 g of cellulose and 0.5 g of CMC under stirring at 3000 r/min and -12.5°C . Then, a designed amount of pre-hydrolyzed TBT solution was added into the above cellulose solution under stirring. The resultant transparent solution was degassed by centrifuging at 8000 r/min and -5°C for 5 min, poured into a mold to allow gelation at room temperature for 12 h, and then immersed in the deionized water to remove NaOH and urea until the wash water was neutral. Consequently, the opaque cellulose hydrogel containing in-situ synthesis of TiO_2 was obtained. The addition of pre-hydrolyzed TBT solution was designed to be 0, 2.5, 5.0, 7.5 and 10.0 mL, and the corresponding hydrogels were marked as TiO_2 -0, TiO_2 -2.5, TiO_2 -5.0, TiO_2 -7.5, and TiO_2 -10.0 nanocomposite hydrogels, respectively.

In-situ mineralization was utilized to load Fe_3O_4 into the as-prepared TiO_2 nanocomposite hydrogels. First, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ and $\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$ were dissolved in deionized water. The molar ratio of Fe^{2+} and Fe^{3+} was 1/2 and the concentration of Fe^{2+} was 0.1 mol/L. Then, the as-prepared TiO_2 nanocomposite hydrogels were soaked in the mixture solution of FeCl_2 and FeCl_3 aqueous solution for 2 days and further soaked in 25 wt% ammonia aqueous solution for another 2 days. Finally, the cellulose-based nanocomposite hydrogels supported $\text{TiO}_2/\text{Fe}_3\text{O}_4$ dual-component catalysts were washed off any excess ammonia for further characterization and tests.

Characterization

Compressive behavior of TiO_2 nanocomposite hydrogels was measured using a UTM2503 universal testing machine (Shenzhen Suns Technology Stock Co. Ltd., China) at room temperature with a compression speed of 5 mm/min, and the samples were cut into a cylinder with a diameter of 1 cm and a height of 1 cm. In order

to characterize the chemical components and microscopic structure of nanocomposite hydrogels, the xerogels obtained by oven-drying hydrogel samples for 24 h at 80 °C, were ground and used for Fourier transform infrared spectroscopy (FTIR), X-ray diffraction (XRD) and X-ray photoelectron spectroscopy (XPS) characterizations and the freezing-dried aerogels were used for thermogravimetric analyses (TGA), field emission scanning electron microscopies (FESEM), energy-dispersive X-ray spectroscopy (EDS) and transmission electron microscope (TEM) measurements to maintain the original dispersion state of catalysts. FTIR characterization was carried out on a Nicolet 6700 spectrometer (ThermoFisher Scientific, USA) in the transmission mode. XRD curves of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel and cellulose/CMC hydrogel were collected in the 2θ range of $10^\circ \sim 80^\circ$ using a Rigaku Ultima IV diffractometer (Rigaku, Japan) at 40 kV and 100 mA. TGA was conducted on a TG209F1 instrument (NETZSCH, Germany) at a heating rate of $10^\circ\text{C}/\text{min}$ from 30 to 800°C in air flow. XPS was taken by AXIS Ultra DLD (Kratos, Japan). The porous morphology was observed by using FESEM (FEI Inspect-F, Finland) at an accelerating voltage of 5 kV. All of the samples for SEM observation were cryo-fractured in liquid nitrogen to maintain their original morphology. The elemental analysis was conducted with EDS (JSM-7500F Electron Optics Laboratory, Japan). The cross-sectional TEM images of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel were observed on a HT7800 electron microscope (HITACHI, Japan) using an accelerating voltage of 200 kV. The electron spin resonance (ESR) signals of 5,5-dimethyl-1-pyrroline *N*-oxide (DMPO) spin-trapped radicals were recorded at ~ 9.8 GHz microwave frequency and 100 kHz magnetic field modulation frequency with an ESR spectrometer (EMX Plus, Bruker, USA).

Absorption, degradation and desorption

The as-prepared $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogels were cut into small cube with a side-length of ~ 1 mm for absorption, degradation and desorption tests (Fig. S1). To the absorption experiment, the container was covered by aluminium foil, and 0.50 g of nanocomposite hydrogel sample was added into 50 mL methylene blue (MB) aqueous solution ($c_0 = 20$ mg/L) under stirring. And hydrochloric acid

was used to regulate the pH of the solution, the amount of H_2O_2 was controlled from 0 to 50 mmol. The concentration of MB was monitored in real time by taking 3 mL solution every 10 min for absorbance measurement, using a UV–vis spectrophotometer Lambda 750S (Perkin Elmer, USA). In the degradation experiment, the container, MB aqueous solution and hydrogels were exposed to ultraviolet light (300 W UV light, OSRAM), and the light source was placed 15 cm away from the container. As for desorption test, the hydrogels were firstly added into MB aqueous solution ($c_0 = 20$ mg/L, 50 mL) for 60 min, then transferred to an acid solution with the equal volume (pH = 4) (Lin et al. 2019). The concentration of MB was monitored as same as that in the absorption and degradation tests. To the cyclic tests, the hydrogel was simply taken out after absorption or degradation and directly put into a new MB solution for the next round test without further treatment.

Results and discussion

Morphology and structure analysis

For the purpose of developing highly dispersed $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel, we designed the hydrogel structure according to the following two principles. First, the abundant functional groups of $-\text{OH}$ and $-\text{COOH}$ were utilized as active sites for in-situ synthesis or mineralization of TiO_2 and Fe_3O_4 , achieving highly dispersed dual-component catalysts supported in the cellulose hydrogel. Second, the interpenetrating cellulose network containing $-\text{OH}$ and $-\text{COOH}$ provided a good platform for the sufficient contact between TiO_2 and Fe_3O_4 , thus, synergistically promoting the degradation of contaminants in the waste water via photo-Fenton reaction. Figure 1a schematically outlines the procedure for the preparation of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel. Initially, TBT was stabilized by mixing with TEA and then added into cellulose/CMC solution, in which nano-scaled TiO_2 could be slowly obtained in the NaOH/urea aqueous system. Interestingly, the addition of TiO_2 significantly facilitated the gelation process that was completed within 12 h at room temperature while there was no sign of gelation for cellulose/CMC solution without TBT (Fig. 1b). This indicates that TiO_2 participates in the reconstruction of

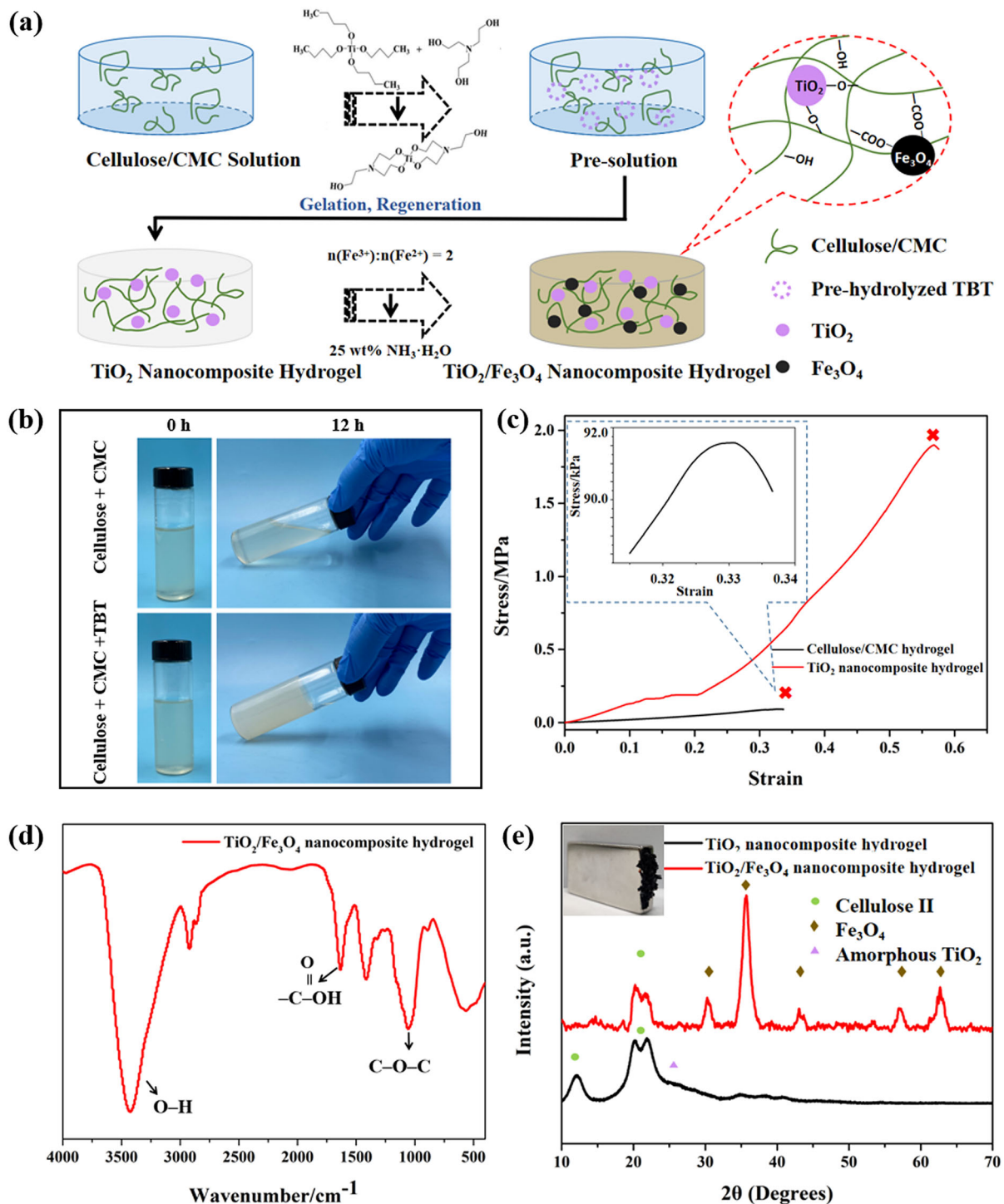


Fig. 1 **a** Schematic graph of preparing $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel; **b** the photographs of precursor solution with and without TBT in 0 h and after 12 h; **c** the compressive test of cellulose/CMC and their TiO_2 nanocomposite hydrogel; **d** FTIR

spectrum of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel; **e** XRD spectra of TiO_2 and $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogels. The inset is a photograph of xerogel that attached on the magnet

hydrogen bonding network, which is further confirmed by the remarkably enhanced compressive strength shown in Fig. 1c. The enhanced mechanical properties by in-situ synthesis are conducive to supporting the cyclic catalytic degradation of cellulose-based nanocomposite hydrogel for the water remediation. The combination of cellulose and CMC in hydrogel structure is simply verified by the appearance of the peak of $-\text{COOH}$ at 1636 cm^{-1} in FTIR spectrum (Fig. 1d), providing absorption sites for ferric and ferrous ions (Chang et al. 2011). Consequently, the $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel was obtained via in-situ mineralization with the assistant of ammonia aqueous solution. Figure 1e represents the XRD curves of both TiO_2 and $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogels. The diffraction peaks at $2\theta = 12^\circ$, 20° and 22° are signed to (1–10), (110) and (200) planes of cellulose II crystal (Qi et al. 2009; Zhang et al. 2019). Only a broad peak in the range of $24^\circ \sim 50^\circ$ is observed for TiO_2 nanocomposite hydrogels, implying the amorphous structure of in-situ synthesized TiO_2 . As for $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel, it is vividly seen from the inset in Fig. 1e that the $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite xerogel can be easily attached on the magnet due to the presence of magnetic ferric oxide. The characteristic diffraction peaks of Fe_3O_4 are located at 30.32° , 35.73° , 43.13° , 57.07° , 62.76° in the XRD curve according to the card JCPDS (19-0629) (El-Desoky et al. 2020). These results confirm that in-situ synthesis or mineralization of TiO_2 and Fe_3O_4 had been successfully achieved and the dual-component catalysts were purposefully loaded into the molecular chains network of cellulose hydrogels.

To better understand the structural change of cellulose-based nanocomposite hydrogels by introducing TiO_2 and Fe_3O_4 , XPS analysis of cellulose/CMC, TiO_2 and $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogels were conducted. As depicted in Fig. 2a, the high-resolution signals of C 1s elements for cellulose/CMC hydrogel can be deconvoluted into three peaks at 248.7 eV (C–C/C–H), 286.3 eV (C–C–O) and 287.5 eV (C–C = O/O–C–O), corresponding to the carbon atoms of $-\text{CH}_2-$, C–OH and C–OOH/O–C–O, respectively (Zhang et al. 2014). After introducing TiO_2 , the peaks of C–OH and C–OOH for cellulose and CMC have a shift of 0.1 eV and 0.3 eV to higher region, probably originating from the crosslinking between cellulose and TiO_2 . Figure 2b shows the O 1s spectra of cellulose/CMC, TiO_2 and $\text{TiO}_2/\text{Fe}_3\text{O}_4$

nanocomposite hydrogels. Obviously, only oxygen in organic functional groups is observed at 532.6 eV for cellulose/CMC hydrogel and the peak areas of lattice oxygen in nanocomposite hydrogels increased from 0 to 7.2% and then to 9.3% by introducing TiO_2 and Fe_3O_4 (Li et al. 2017). In addition, the new peak of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel at 534.1 eV could be attributed to the redistribution of charge density for carboxyl group during in-situ mineralization of Fe_3O_4 . Thus, it is reasonable to speculate that TiO_2 and Fe_3O_4 were successfully incorporated into the hydrogen bonding network of cellulose hydrogels. Furthermore, Fig. 2c displays the typical peaks of TiO_2 at 458.2 and 463.9 eV with a doublet splitting of 5.7 eV , corresponding to $\text{Ti}^{4+} 2p_{3/2}$ and $\text{Ti}^{4+} 2p_{1/2}$, respectively (Jatoi et al. 2019). It is worth noting that the peaks located at 456.9 and 460.7 eV are related to $\text{Ti}^{3+} 2p_{3/2}$ and $\text{Ti}^{3+} 2p_{1/2}$, exhibiting an increased area from 9.0 to 16.7% by introducing Fe_3O_4 . This is indicative of a strong interaction between TiO_2 and Fe_3O_4 , which allows the as-prepared nanocomposite hydrogels to hold great potential as highly efficient catalytic degradation system for the removal of aqueous contaminants via photo-Fenton reaction (Salamat et al. 2017).

The dispersion morphology of TiO_2 and Fe_3O_4 was determined by using SEM/EDS and TEM characterizations. As illustrated in Fig. 3a, b, SEM photograph and corresponding EDS mapping of cross section of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel confirm that TiO_2 and Fe_3O_4 are highly dispersed into the cellulose matrix without any visible aggregates. Furthermore, two kinds of nanoparticles with tubular and granular morphology can be observed in the typical TEM image in Fig. 3c, d. The tubular morphology is demonstrated to be assigned as TiO_2 , as shown in Fig. S2. And the granular nanoparticle had the similar morphology with the other works of in-situ mineralized Fe_3O_4 hydrogel (Ray et al. 2017; Konwar et al. 2019). It is reasonable to conclude that tubular TiO_2 and granular Fe_3O_4 are highly dispersed and fully contacted each other at the nano-scale level, which is the essential prerequisite of an efficient photo-Fenton reaction for pollutant degradation.

The effect of TiO_2 on the crosslinking network of nanocomposite hydrogels was further studied by observing the porous morphology of their corresponding aerogels. As illustrated in Fig. 4a–e, cellulose/CMC hydrogel displays an open, highly porous

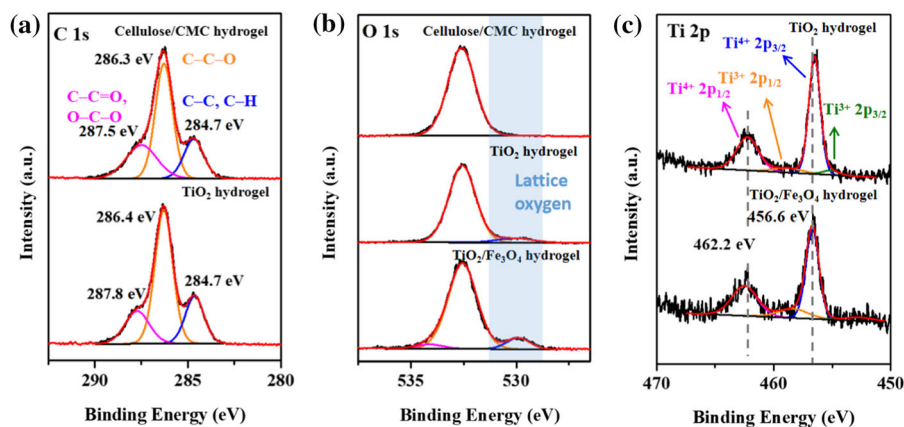
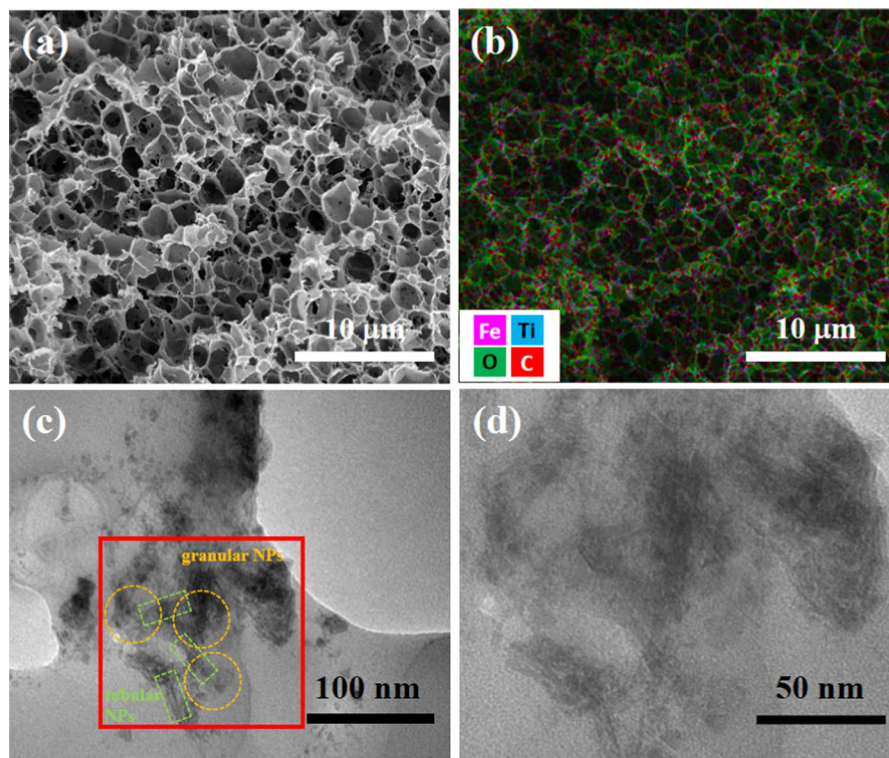


Fig. 2 XPS spectra of cellulose/CMC, TiO_2 and $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogels: **a** C 1s; **b** O 1s; **c** Ti 2p

Fig. 3 **a** Typical SEM images and **b** EDS elemental mapping of the cross section of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel; **c** TEM images of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel with low (**c**) and **d** high magnification



structure, where the major pore size is roughly estimated to be several microns in diameter and a mass of secondary pores with diameters ranging from 100 nm to 1 μm on the micro-sized pore walls. The inclusion of TiO_2 has a significant impact on the porous morphology, exhibiting a decreased pore size and increased network density with the increase of TiO_2 concentration. Besides, TGA curves can be used to roughly determine the concentration of TiO_2 and

Fe_3O_4 in the nanocomposite hydrogels by considering the residues at 800 $^\circ\text{C}$, as shown in Fig. S3. We can also discern from Fig. 4f that there is a gradual increase in the TiO_2 content with increasing the dosage of pre-hydrolyzed TBT and the TiO_2 content performs well in terms of controllability with a good linearity of $R^2 = 0.999$. These results indicate that the porous morphology and network density largely depend on the TiO_2 content and the loaded TiO_2

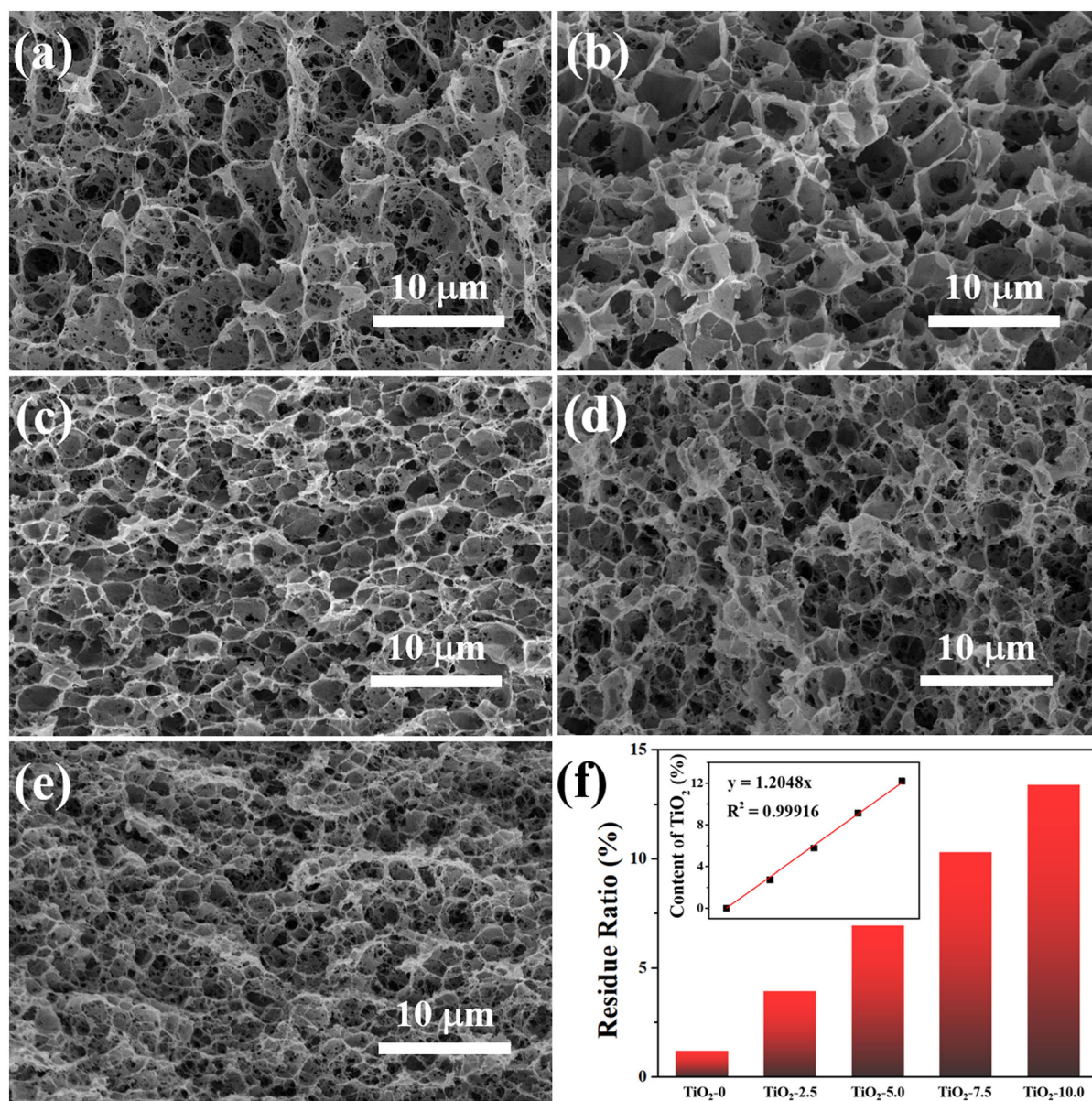


Fig. 4 Typical SEM images of cellulose hydrogel after freeze-drying without pre-hydrolyzed TBT (a); b TiO₂-2.5; c TiO₂-5.0; d TiO₂-7.5; e TiO₂-10.0 hydrogels; f the residue ratio of the corresponding aerogels with different concentrations of pre-hydrolyzed TBT

always plays as the crosslinking site rather than aggregate in the nanocomposite hydrogels.

Absorption and degradation properties

For evaluating the absorption and the catalytic activity of TiO₂/Fe₃O₄ nanocomposite hydrogels, MB was chosen as a model organic dye contaminant and the detailed absorption and degradation test were

elaborated in the Experimental Section. First, the degradation activity of both Fenton and photo-Fenton reactions greatly depends on the H₂O₂ concentration, as shown in Fig. S4. The optimal dosage of H₂O₂ is 20 mmol. Figure 5 presents the dependence of degradation ability for TiO₂/Fe₃O₄ nanocomposite hydrogels on the content of TiO₂. The degradation condition was controlled at pH = 4 and 20 mmol H₂O₂. It is observed that c/c_0 values gradually decrease with

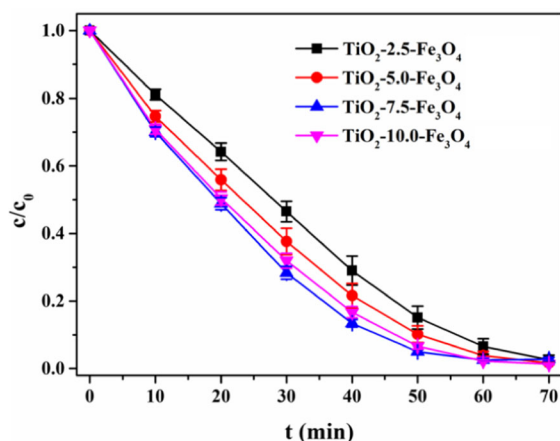


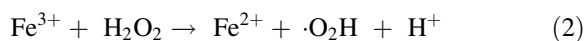
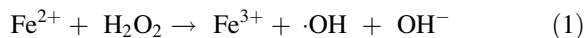
Fig. 5 Photo-Fenton reaction for degradation of MB by TiO₂/Fe₃O₄ nanocomposite hydrogels with different content of TiO₂ ($c_0 = 20$ mg/L; 20 mmol H₂O₂; pH = 4)

increasing degradation time and the corresponding MB degradation efficiency is as high as $\sim 97.0\%$ for all the groups in 70 min. Obviously, the degradation activity of TiO₂/Fe₃O₄ nanocomposite hydrogels largely depends on the concentration of TiO₂. From TiO₂-2.5-Fe₃O₄ to TiO₂-7.5-Fe₃O₄ nanocomposite hydrogels, more loaded TiO₂ means more active site, generating more reactive oxygen species (ROS) for improving the degradation efficiency. Nevertheless, the degradation activity of TiO₂-10.0-Fe₃O₄ nanocomposite hydrogel is inferior to TiO₂-7.5-Fe₃O₄ nanocomposite hydrogel. In this case, although the high content of TiO₂ provides more active site, it results in a denser network confirmed in Fig. 4, which has a negative effect on the internal mass transfer and reduces the degradation efficiency. Thus, the TiO₂-7.5-Fe₃O₄ nanocomposite hydrogel exhibits the highest degradation activity via photo-Fenton reaction, achieving a high removal ratio of 95% in 50 min.

In order to shed light on the degradation mechanism of the as-prepared cellulose-based nanocomposite hydrogels supported TiO₂/Fe₃O₄ dual-component catalysts, the absorption and degradation activity of TiO₂-7.5 and TiO₂-7.5-Fe₃O₄ nanocomposite hydrogels, as well as the controlled samples under different degradation condition were conducted. Some typical degradation curves are listed in Fig. 6a. There is negligible degradation of MB without catalysts in the presence of only H₂O₂ under dark, while the removal ratio can be enhanced to be approximately 51.5% with the assistance of UV light. The degradation efficiency

of TiO₂-7.5-Fe₃O₄ nanocomposite hydrogel via Fenton reaction and TiO₂-7.5 nanocomposite hydrogel via photocatalysis is 56.3% and 95.5% in 70 min, respectively. In this case, the photocatalysis is obviously superior to the advanced oxidation from Fenton reaction in degrading the organic dyes of MB. As expected, it is striking that the removal ratio of TiO₂-7.5-Fe₃O₄ nanocomposite hydrogel rises up to 97.5% via photo-Fenton reaction. Such a high degradation efficiency is highly attractive in the field of hydrogel supported catalyst. Besides, it is worth noting that the removal ratios via absorption for TiO₂-7.5 and TiO₂-7.5-Fe₃O₄ nanocomposite hydrogels are 41.1% and 37.4%, respectively. The reduced absorption efficiency by the addition of Fe₃O₄ can be interpreted on the grounds of the decreased absorption sites, which are occupied by ferric/ferrous ions to achieve denser crosslinking network. Compared with TiO₂-7.5 nanocomposite hydrogel, the inferior absorption but superior degradation of TiO₂-7.5-Fe₃O₄ nanocomposite hydrogel further manifest the superiority of the dual-component catalytic system containing TiO₂ and Fe₃O₄ for the effective removal of organic dyes via photo-Fenton reaction.

This optimal degradation activity of photo-Fenton reaction also reflects in the DMPO spin-trap ESR test that was used to monitor ROS. As depicted in Fig. 6b, it is vividly seen that photo-Fenton reaction displays the strongest signal of $\cdot\text{OH}$ at pH = 4 relative to Fenton reaction and photocatalysis. The phenomenon should be attributed to the following two aspects. First, as reported previously, the electron generated by TiO₂ photocatalysis can accelerate the reduction of Fe³⁺ to Fe²⁺, which is the rate-limiting step of Fenton reaction in Eq. 2 (Zhu et al. 2019).



Second, the lifetime of TiO₂ photocatalysis reaction can be prolonged as the photo-generated hole is more difficult to recombine with the photo-generated electron that has been captured by Fe₃O₄ (Li et al. 2019). Taking the discussion of cellulose hydrogel networks into consideration, we propose the possible photo-Fenton degradation mechanism of TiO₂/Fe₃O₄ nanocomposite hydrogel in Fig. 6b. First, the three-dimensional network with high water retention formed by intra- and inter-molecular hydrogen bonding leads

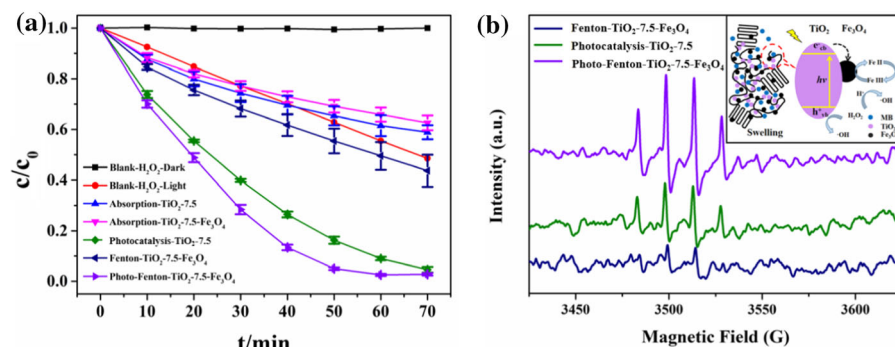


Fig. 6 a Absorption and catalytic degradation curves of blank sample with only H_2O_2 , TiO_2 -7.5 and TiO_2 -7.5- Fe_3O_4 nanocomposite hydrogels for Fenton, photocatalyst and photo-Fenton reaction (pH = 4, and 20 mmol H_2O_2 for degradation);

b DMPO spin-trapping ESR spectra of $\cdot OH$ radicals at pH = 4 for different reactions. The inset is the possible photo-Fenton degradation mechanism of TiO_2/Fe_3O_4 nanocomposite hydrogel

to good internal mass transfer, ensuring effective contact between pollutant and catalysts. Moreover, the interpenetrating cellulose network containing $-OH$ and $-COOH$ provides a good platform for achieving highly dispersed TiO_2 and Fe_3O_4 with sufficient contact at nano-scale, which is the key point to obtain an efficient photo-Fenton reaction. Thus, the as-prepared TiO_2/Fe_3O_4 nanocomposite hydrogel holds great potential or value as a high effective catalyst for water remediation.

The superiority of TiO_2/Fe_3O_4 nanocomposite hydrogel for water remediation

Applied pH range for photo-Fenton degradation

The initial value of pH is considered as one of the most important factors for Fenton-like reaction. Figure 7 summarizes the removal ratio of absorption and degradation in the pH range of 4 ~ 7 and the corresponding curves of c/c_0 as a function of degradation time are shown in Fig. S5. Since the carrier of catalyst in the current study was designed as ionic hydrogel which acts as swelling or shrinking at different pH (Chang et al. 2011). The initial value of pH is an important parameter to determine the internal mass transfer. Predictably, as listed in Fig. S5, the removal ratio of absorption decreases from 90 to 34% by increasing the acidity of MB aqueous solution because the shrinking of hydrogels at acidic condition gives rise to the denser network and decreases the absorption site. Interestingly, the removal ratios in the degradation test maintain over 97.5% in the broad

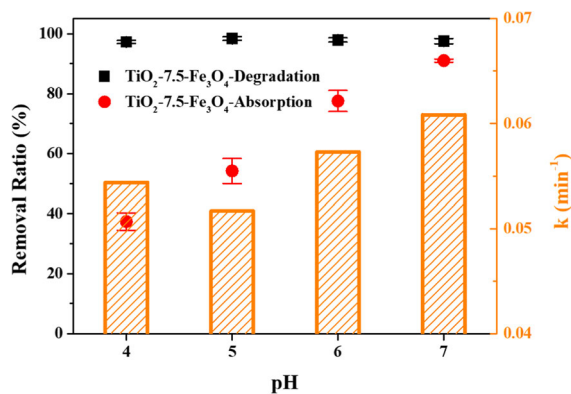


Fig. 7 The removal ratio of absorption (black square) and degradation (red circle) for TiO_2 -7.5- Fe_3O_4 hydrogel in 70 min as well as the degradation rate at different pH (orange histogram)

initial values of pH from 4 to 7. It is also worth to mention that the rate of degradation even increased from pH = 5 to 7, achieving the highest value of 0.061 min⁻¹ at pH = 7. It is well-known that the pH value is commonly limited in the range of 2.0 ~ 4.0 for effective Fenton oxidation, and the neutral solution is not a good condition for both Fenton and photocatalysis reaction. In order to Figure out whether absorption or degradation is the main contributor to the excellent degradation performance, the desorption test was performed immediately after degradation under neutral condition for 70 min. As shown in Fig. S6, there is no any trace of MB in the desorption test, confirming that the removal process under the neutral condition is overwhelmingly ascribed to degradation rather than absorption. The better

swelling property of anionic cellulose hydrogel under neutral condition endows $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel with loose network and good internal mass transfer, which can effectively improve the degradation efficiency of photo-Fenton reaction. Another benefit of the good degradation performance at neutral condition is to broaden the applied pH range, avoiding the high consumption of acid that accounts for a large proportion of cost.

Stability and recyclability of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel

The potential reusability of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel is an essential prerequisite for practical application. In view of good degradation performance of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel under neutral condition, the cyclic absorption and degradation were also carried out for 5 times at $\text{pH} = 7$ to evaluate its stability and reversibility. As shown in Fig. 8, the remove ratio of absorption decreased from 89.8 to 46.2% because the absorbed MB in the former cycle occupied the absorption site, while the removal ratio of degradation in 70 min for each cycle reached to 97.5%. Anyhow, the nanocomposite hydrogel in this case has an edge over the previous works in reusability test (Table S1). For the hydrogel supported catalysts, it usually had a decrease of removal ratio because of the unstable network between catalysts and hydrogel during free radical reaction. In our case, the excellent reversibility of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel mainly arises from its unique structure design including the robust three-dimensional cellulose network constructed by a large amount of intra- and inter-molecular hydrogen bonding and firm anchoring of TiO_2 and Fe_3O_4 on the molecular chains of cellulose, effectively avoiding the destabilization of hydrogel during degradation.

Safe management in degradation

Apart from the stability in service, it is also a key point if the carrier supported catalysts could safely and tenaciously handle the pollutant, especially for the water soluble one. Thus, the desorption tests were taken under acidic condition for determine the safe management ability of designed $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel during degradation. As shown in Fig. 9, cellulose/CMC and $\text{TiO}_2/\text{Fe}_3\text{O}_4$

nanocomposite hydrogels absorbed 71.5% and 93.1% of the initial MB in 70 min under neutral condition. Nevertheless, the leakage of MB during desorption tests for them were 47.1% and 28.8% respectively. Compared with cellulose/CMC hydrogel, the higher absorption rate but lower desorption rate of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel should be attributed to the loaded nanoparticles that serve as absorption sites. Moreover, the leaked MB can be quickly removed with the assistant of H_2O_2 and UV light. To a certain extent, this designed nanocomposite hydrogel is like a reactor where photo-Fenton reaction takes place and nearly no organic MB molecule is leaked. With the highly active degradation and the excellent ability of tenaciously holding pollutant, the as-prepared nanocomposite hydrogel supported catalyst has a great potential for safe handling of waste water.

Conclusion

In summary, a novel cellulose-based nanocomposite hydrogel containing highly dispersed dual-components catalysts of TiO_2 and Fe_3O_4 was successfully fabricated by in-situ synthesis and in-situ mineralization, which was confirmed by XRD curves, XPS spectra, EDS elemental mapping and TEM images. The TiO_2 was found to participate in the reconstruction of hydrogen bonding network, which was reflected by accelerated gelation process, enhanced compressive strength, as well as the decreased pore size of cellulose-based nanocomposite aerogels. The Fe_3O_4 was anchored onto the cellulose chains, exhibiting typical magnetic feature of cellulose-based xerogel. As a result, compared with Fenton reaction and photocatalysis, the as-prepared $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel was highly efficient for the removal of organic dyes via photo-Fenton reaction and the corresponding removal ratio was as high as 97.5% in 60 min. More importantly, the degradation efficiency was maintained from $\text{pH} = 4 \sim 7$ and there was no decrease after 5 cycles of degradation. Besides, the nanocomposite hydrogels could firmly hold the contaminants during degradation and nearly no organic MB molecule was leaked. The methodology proposed here provides a new insight in preparing multi-component catalysts in nanocomposite

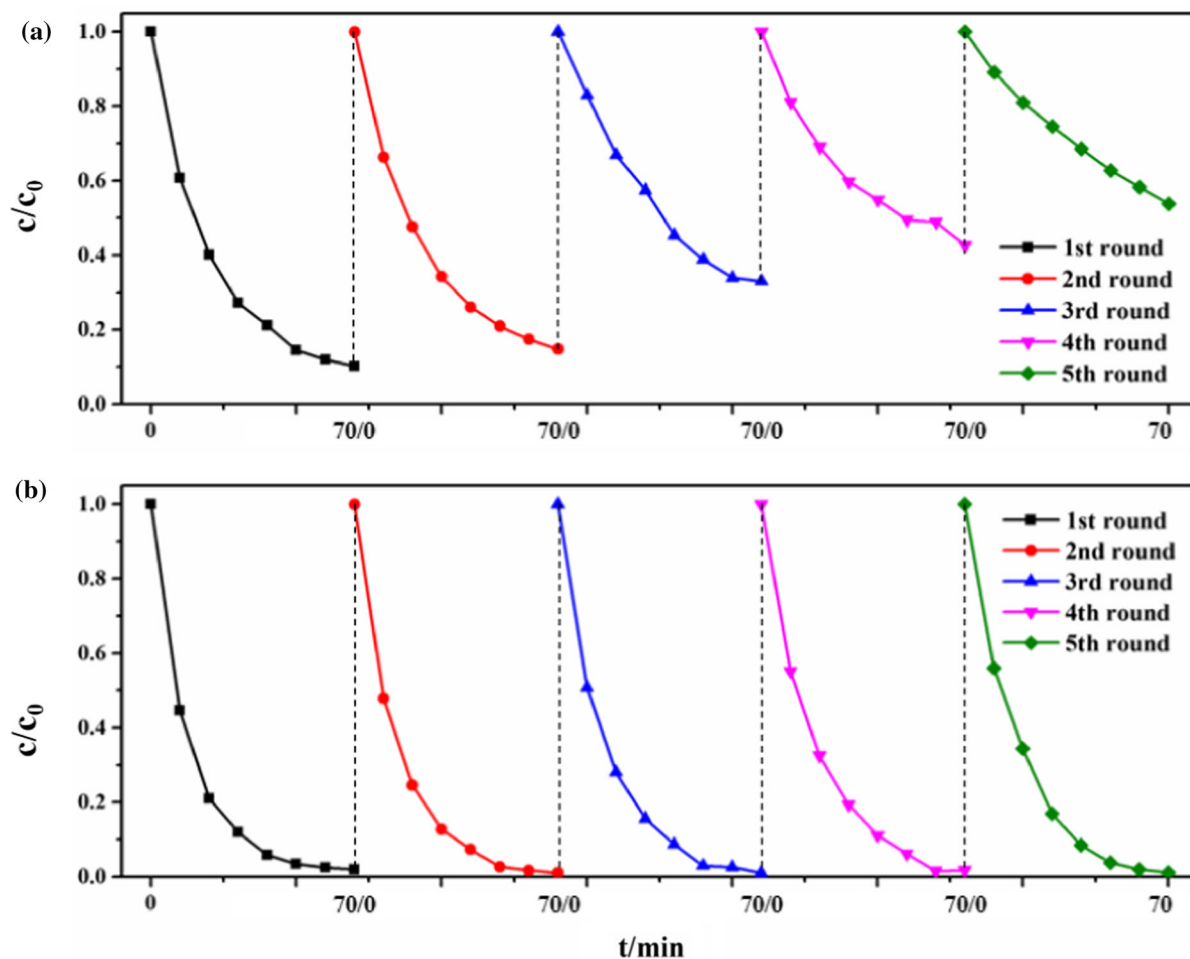


Fig. 8 Cyclic tests of $\text{TiO}_2/\text{Fe}_3\text{O}_4$ nanocomposite hydrogel for absorption (a) and photo-Fenton degradation (b) at $\text{pH} = 7$

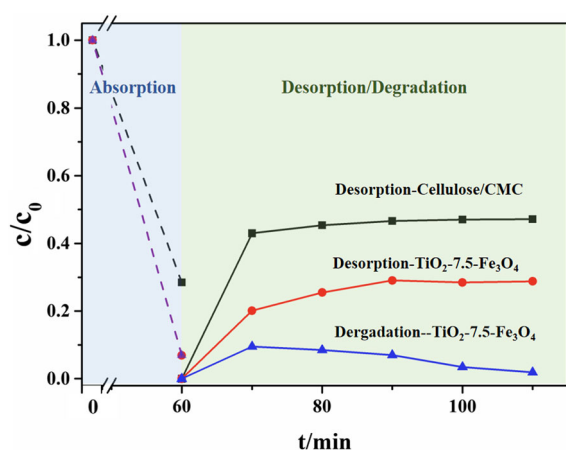


Fig. 9 The desorption and degradation test for the nanocomposite hydrogels ($c_0 = 20 \text{ mg/L}$)

hydrogels for efficient, stable, and safe treatment of water pollutants.

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Declarations

Conflict of interest There are no conflicts to declare.

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